Revised Dynamic Emotion Simulation Benchmark, v2

May 2, 2026

1 Purpose

This report summarizes the revised benchmark evidence for the Behavior Research Methods revision. The first aim is to compare dimensional recovery methods under the current dynamic latent oscillator simulation design. The second aim is to show whether the calibrated heterogeneous simulation produces emotion-score properties that are close enough to the OpenAI-large empirical target to support a cautious empirical-realism claim. This v2 report replaces the earlier time-series comparison display with metric tables, distributional summaries, and simulated examples from the benchmark scenarios.

2 Benchmark Comparison

The main benchmark treats DynEGA and GraphicalVAR as the full-grid comparison. Both methods were evaluated on 1,920 rows formed by crossing the four expression regimes, two to four latent dimensions, sample size, time-series length, number of observed variables per dimension, and error level. GIMME is not treated as a full-grid peer because the full design was computationally unstable; it is reported only as a bounded feasibility comparator on 120 rows.

Table 1: Method-level benchmark overview. Exact recovery is calculated over all benchmark rows, so timeouts count against recovery. Runtime is the mean wall-clock time per benchmark row.

Method	Scope	Runs	Success	Timeout	Exact	Cor.	NRMSE	Runtime
DynEGA	Full grid	1920	100.0%	0.0%	97.4%	0.654	0.867	3.6 s
GraphicalVAR	Full grid	1920	97.5%	2.5%	87.3%	0.649	0.865	191.4 s
GIMME	Bounded feasibility	120	17.5%	80.0%	15.8%	0.614	0.944	364.3 s

DynEGA is the strongest default method in this benchmark. It completed all full-grid rows, recovered the correct dimensionality in 97.4% of rows, and had nearly identical score-recovery error to GraphicalVAR while requiring far less runtime. GraphicalVAR remains a useful structural comparator because its correlation and NRMSE summaries are similar to DynEGA, but its recovery is less stable in the harder four-dimensional conditions and its average runtime is about fifty times larger. GIMME provides useful feasibility evidence, but its all-row result is dominated by timeouts and should be presented separately from the main benchmark comparison.

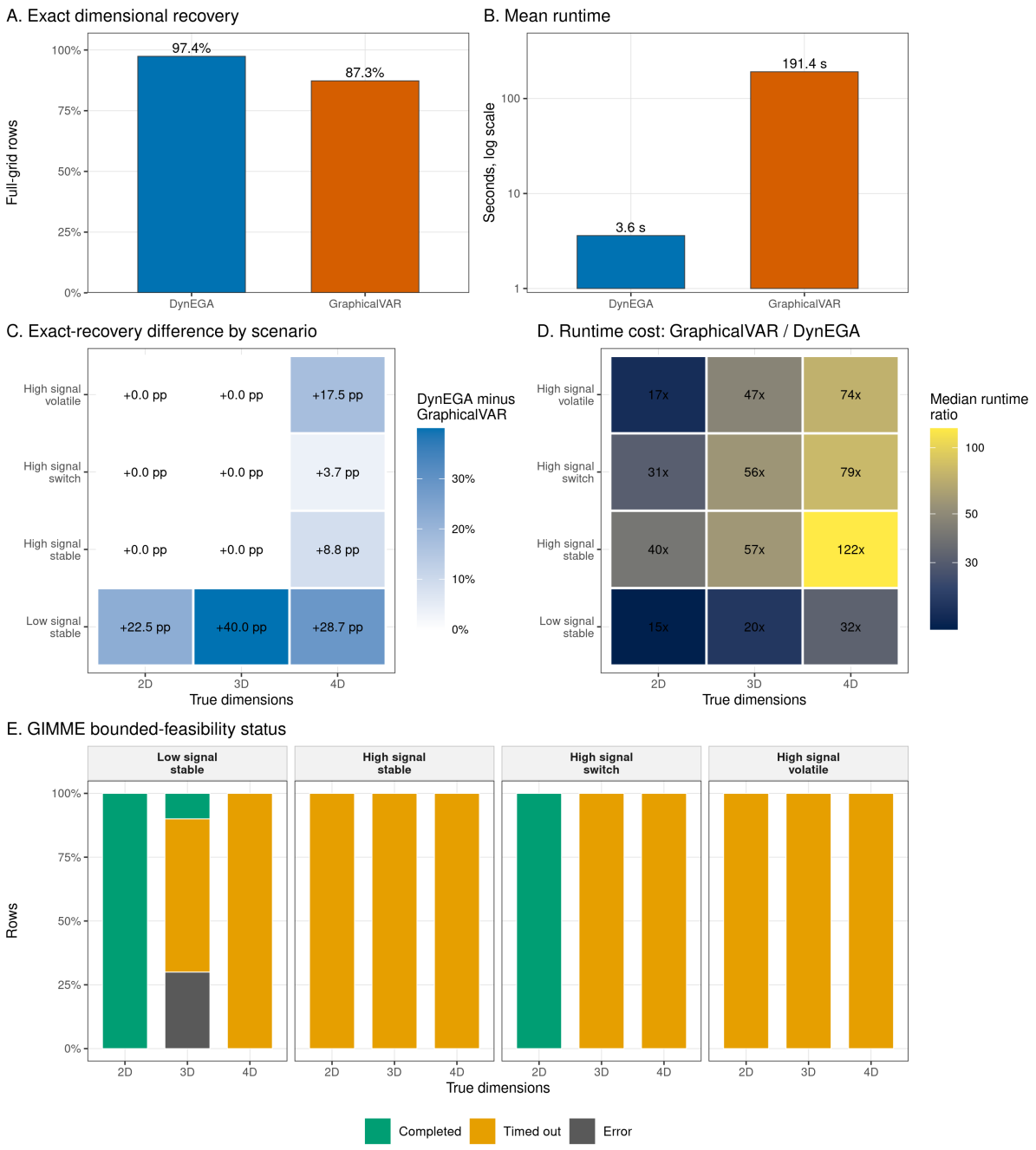


Figure 1: Clean benchmark-difference summary. Panels A and B show the method-level full-grid difference in exact dimensional recovery and runtime. Panel C shows where DynEGA gains exact-recovery percentage points over GraphicalVAR across scenario regimes and true dimensionality. Panel D shows the median runtime ratio for GraphicalVAR relative to DynEGA. Panel E reports GIMME only in its bounded feasibility role.

3 OpenAI-large Empirical-Realism Check

The OpenAI-large comparison is summarized with distributional metrics rather than a direct visual comparison of observed and synthetic trajectories. This is the cleaner target for the revision because the empirical question is whether the simulator reproduces broad time-series properties, not whether a synthetic trajectory should visually resemble any particular observed clip. The table reports medians for the empirical OpenAI-large series and the heterogeneous DLO simulation, together with scaled median differences and scaled distributional distances.

Table 2: OpenAI-large versus heterogeneous DLO metric comparison. Median differences and distances are scaled by empirical dispersion, so smaller values indicate a closer empirical-realism match.

Family	Metric	OpenAI-large	DLO sim.	Median diff.	Distance
Latent	Sample entropy	0.427	0.195	1.08	0.94
Latent	Lag-1 ACF	0.567	0.820	0.90	0.87
Latent	Standard deviation	0.321	1.704	3.99	4.16
Latent	Total variation	0.179	0.506	1.81	1.65
Latent	Zero-crossing rate	0.214	0.081	0.96	0.90
Raw	Sample entropy	0.410	0.155	1.08	0.86
Raw	Lag-1 ACF	0.542	0.533	0.03	0.04
Raw	Standard deviation	0.020	0.077	1.11	1.17
Raw	Total variation	0.011	0.038	0.89	0.85
Raw	Zero-crossing rate	0.225	0.160	0.42	0.33

The empirical-realism evidence is mixed but useful. Raw lag-1 autocorrelation is closely matched, and raw zero-crossing rate is reasonably close. The simulation departs more strongly on sample entropy, standard deviation, and total variation, with the latent score space showing larger smoothness and amplitude mismatches. The appropriate manuscript claim is therefore not full distributional equivalence. The better claim is that the calibrated DLO panels capture some broad dependence properties of the OpenAI-large series while still showing measurable departures in local irregularity and movement amplitude.

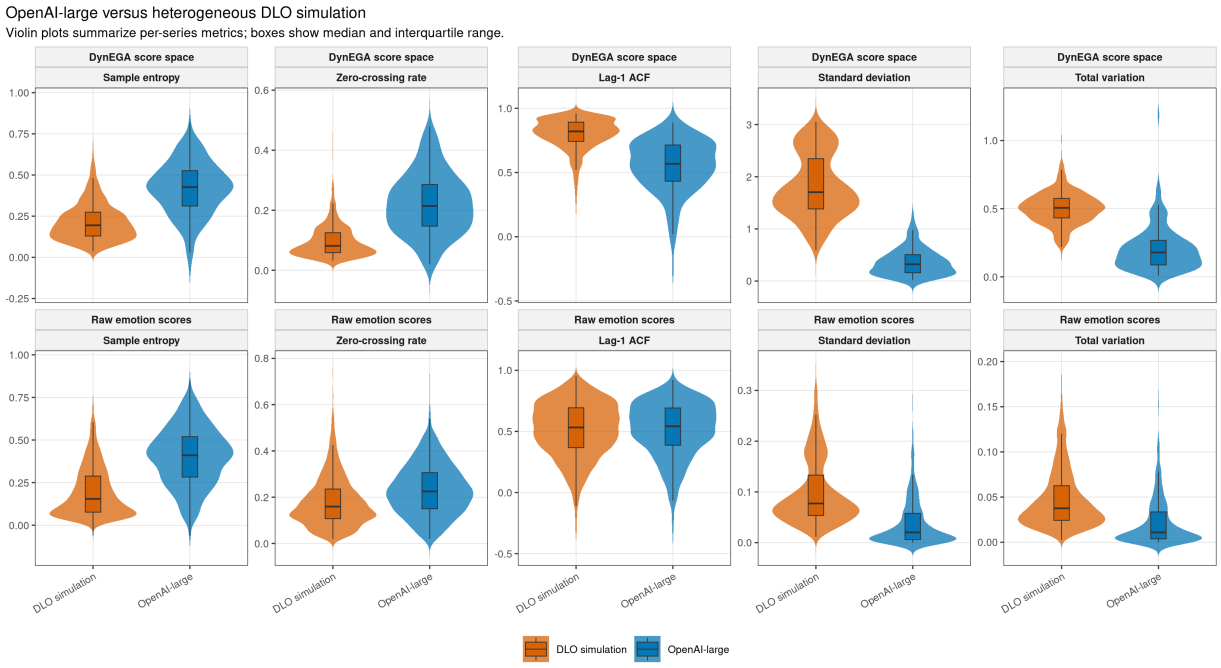


Figure 2: OpenAI-large versus heterogeneous DLO metric distributions. Violin plots show per-series metric distributions for raw emotion-score series and the DynEGA score space; embedded boxplots show medians and interquartile ranges.

4 Benchmark Scenario Examples

The report now uses simulated examples only to illustrate the benchmark scenario families. These examples are not empirical comparisons. They show one simulated two-dimensional video from each expression regime used in the benchmark, making the distinction between low-signal stable, high-signal stable, switching, and volatile switching scenarios visible without implying trajectory-level correspondence to OpenAI-large clips.

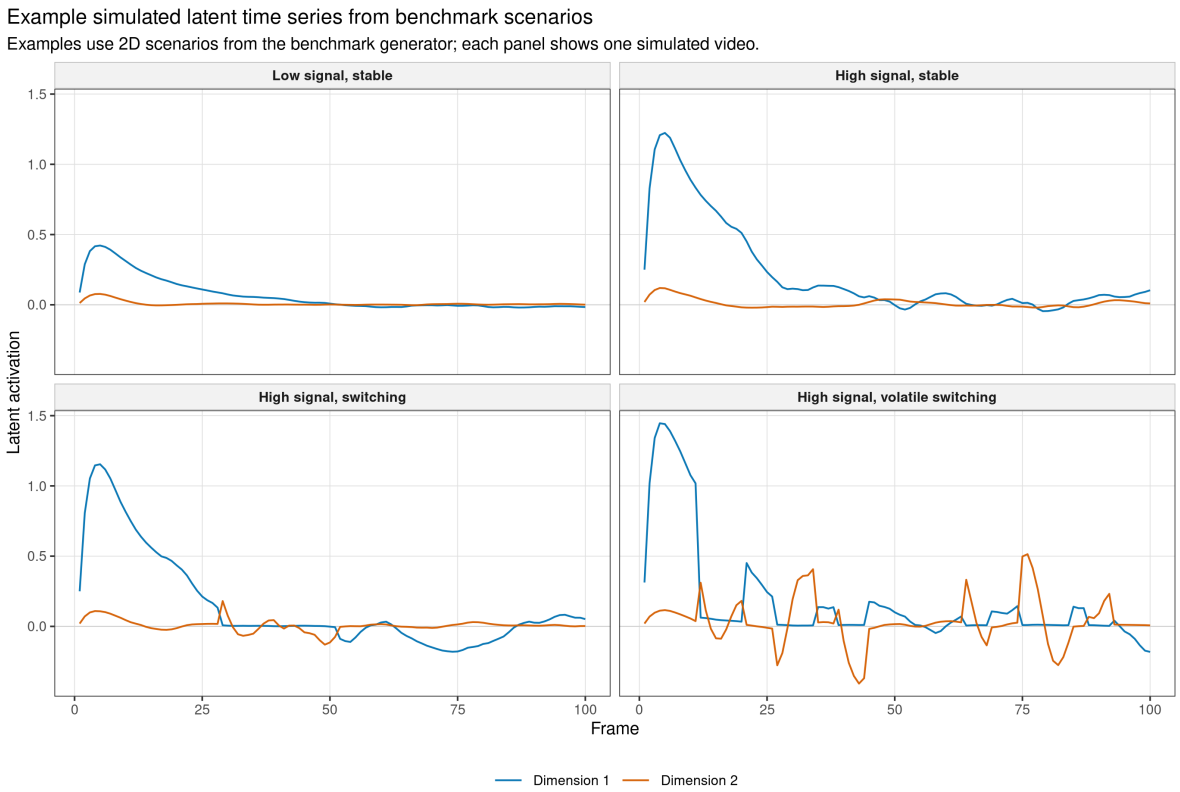


Figure 3: Example simulated latent time series from the benchmark generator. Each panel shows one two-dimensional simulated video from a benchmark scenario mode. The figure illustrates the simulation regimes used for method recovery rather than comparing simulated and empirical trajectories.

5 Revision Implications

The revised manuscript can present DynEGA versus GraphicalVAR as the primary benchmark result and reserve GIMME for a bounded feasibility subsection with explicit timeout accounting. The OpenAI-large comparison should be phrased as a metric-based realism check rather than a visual trajectory match. In writing, the strongest defensible statement is that the current simulator is suitable for controlled dimensional-recovery benchmarking and partially reproduces empirical time-series structure, while remaining imperfect on entropy and amplitude properties.